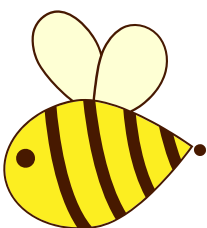


# INTERACTIVE ART STORYBOARDING

## ASSIGNMENT 1 CS6042

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- **Echo Garden: Pollination Playground**  
**Theme: United Nations Sustainable Development**  
**Goal 15 : Life on Land**



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## 1. Introduction

We created a storyboard for non interactive art installation called Echo garden:Pollination Playground. The project focuses on visitor experience and how people might interact with a digital ecosystem and influence virtual plants and pollinators through sketches and flow charts that depicts the systems interactions. Our main goals were to explore the narrative , interactivity and themes in a structured way , analysing how the system would behave and respond.

## 2. Concept Overview

Echo Garden creates a digital world where visitors act like pollinators, leaving virtual pollen trails that make flowers bloom. This experience focuses on:

**Individual impact:** Each visitor contributes to the growth of the digital ecosystem.

**Collective actions:** When multiple visitors interact, larger and more flowers bloom rapidly, showing how working together brings advantage.

**Immersive storytelling:** Visual and audio feedback tells us the health and importance of the ecosystem, guiding visitors throughout the scene. By adding human actions to ecological consequences, this installation highlights Life on Land (SDG 15) and encourages awareness of sustainability

## 3. Storyboard Description

The storyboard consists of six frames that capture the movement of visitors interaction in Echo Garden:



**FRAME 1: EMPTY MEADOW (visitors before approaching the ecosystem)**

Before anyone steps inside, the room is dark and still. The floor shows dry, cracked soil as if it has been waiting a long time for something to change. Alongside, the walls hold faint outlines of flowers without movement. The whole space carries the feeling of a world that could bloom, but hasn't yet. Everything depends on what happens next.

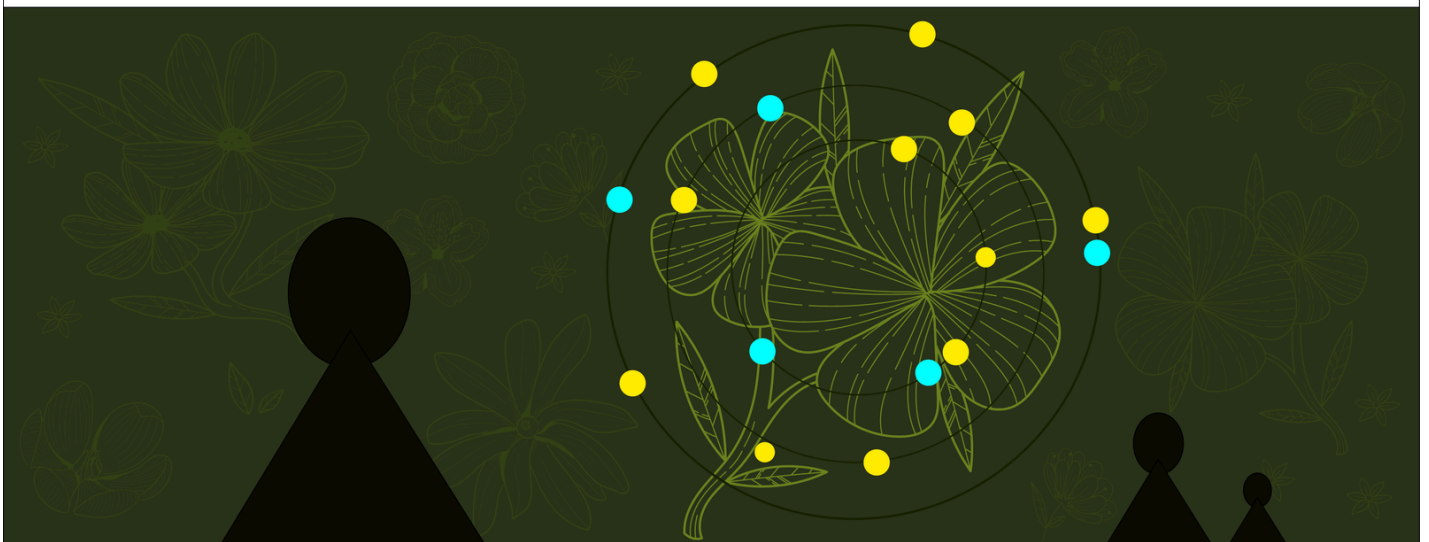
**FRAME 2: STEP INTO LIFE (pollen particles will be generated and carried with each step)**

As the first footstep lands, sensors trigger beneath the floor. Golden pollen particles rise from under each foot, tracing a soft glowing trail with every step. The ecosystem responds directly to the visitor's movement, where they walk life follows.



**FRAME 3: CARRING POLLEN (visitors navigates towards flowers to bloom)**

As visitors walk toward the flower projections on the walls, by following the pollen particles trajectory. Proximity sensing calculates the nearest flower, each step brings the pollen closer. The the floor glows with luminous paths that trace visitors (bees) journey like a digital bee navigating meadow.

**FRAME 4: BLOOM (POLLEN THRESHOLD MET)**

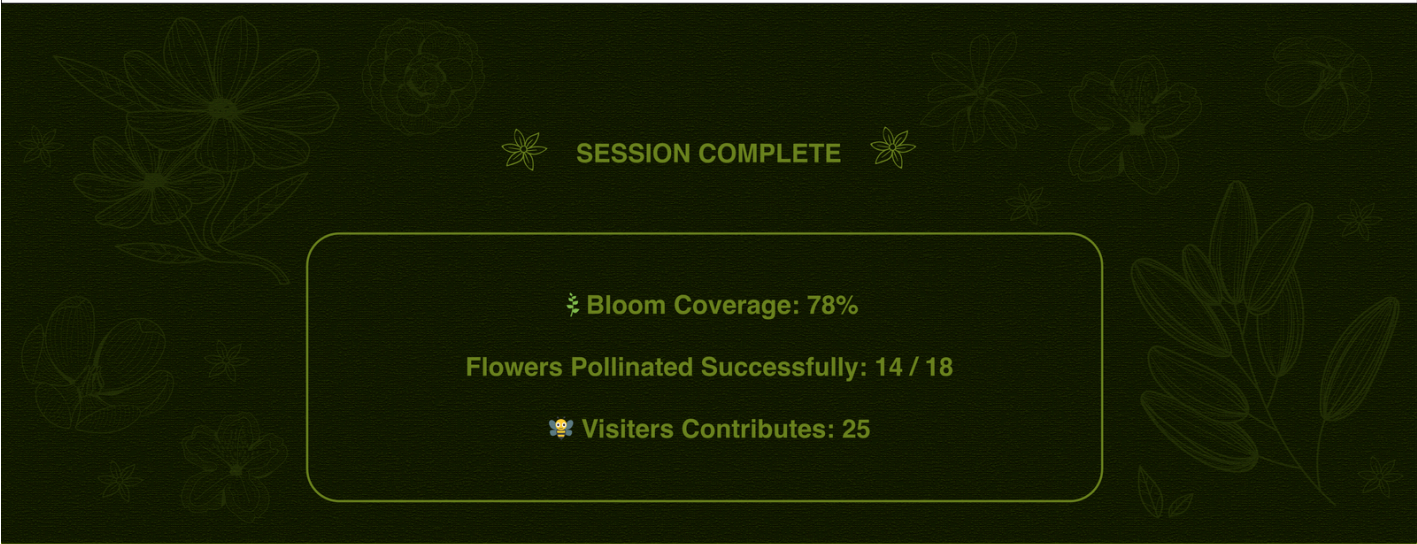
Once the bloom threshold is reached, the flower blooms. Petals unfold across the wall in a burst of colour, releasing a cascade of light particles. A soft chord sounds, layered with birds and wind. The visitor sees the direct result of their movement, one pollen trail and one bloom triggered. A small but visible gain for the ecosystem.

FRAME 5: COLLABRATIVE GROWTH



When multiple visitors move through the space together, the system scales the pollen output for each participant. The individual trails merge into larger, overlapping clouds that reach flowers faster. Blooms spread more quickly across the walls and the soundscape builds with layered harmonics. The meadow visibly thrives in ways that simply aren't possible alone.

FRAME 6: REFLECTION



As visitors prepare to leave, the installation displays a summary of what was collectively achieved such as flowers bloomed, pollen trails mapped, coverage percentage reached. But beyond the numbers, the experience leaves a quiet message: the health of land ecosystems depends on the actions of those within it. Every step taken, every bloom triggered, mirrors the real responsibility we carry toward biodiversity and life on land.



## 4. Conceptual Technology

Even though this project is storyboard based, we thought of potential technologies to see how these can help us design the project and understand how the interaction could work in real life.

### Input:

- Floor sensor mats to detect the movement of visitors
- Raspberry Pi Camera Module for tracking visitors
- Proximity sensors to sense distance to flowers

### Processing / Softwares:

- Unity to create visual effects and particle animations
- TouchDesigner for real time audio visual integration
- Raspberry Pi to manage and process sensor inputs

### Output:

- Wall projections to display flowers and blooming effects
- Floor lighting to visualize pollen trails
- Spatial audio for ambient birdsong, wind, and musical hints
- This section is conceptual, illustrating how the interactive experience could be realized without building a physical installation.

## 5. Interaction Rules and Algorithms

To clearly explain how the interactive system works, we described the rules in pseudocode. This helps show how the system behaves and supports the storyboard, without showing or building a real physical version :

### 1. Pollen Generation

If visitor steps detected:

- Generate pollen particle at step location
- Assign particle color based on visitor



## 2. Flower Bloom Threshold

For each flower:

*If pollen\_count\_near\_flower >= bloom\_threshold:*  
*trigger bloom animation*  
*play audio chord*  
*release light particles*

## 3. Collaborative Scaling

If multiple visitors present:

*total\_pollen = sum(pollen per visitor)*  
*bloom\_speed = scale with total\_pollen*  
*particle\_density = merge trails*

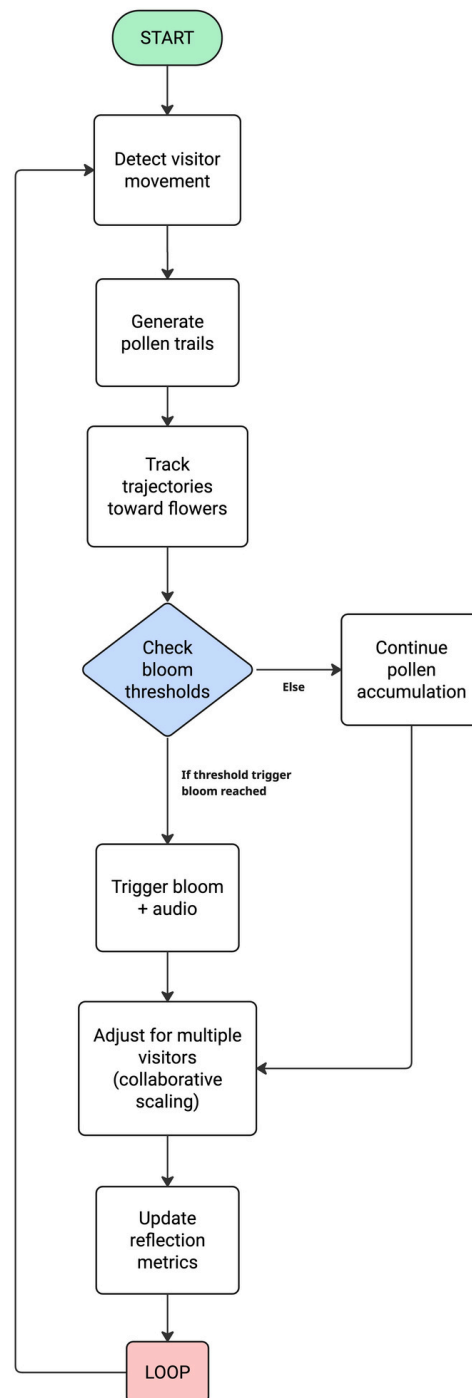
## 4. Reflection Metrics

*bloom\_coverage = (flowers\_bloomed / total\_flowers) \* 100%*  
*visitor\_contributions = count(active participants)*  
*display\_summary(bloom\_coverage, visitor\_contributions)*

These rules are ideas to guide our storyboard. This shows how, what visitors do affects the visuals, sounds, and the digital ecosystem, and how both individual and group actions can change it.

## 6. System Flow/Flow Chart

The below system flow shows how visitors affect the digital ecosystem in Echo Garden :





## **7. Theme Exploration**

Echo Garden illustrates Life on Land (SDG 15) by using pollination as a metaphor for human ecological responsibility. Showing how individual and community activities impact ecosystem health. Transforming abstract sustainability concepts into an immersive narrative experience. Moreover, the storyboard visualise these ideas without the need for a physical exhibit, illustrating the cause and effect links between visitor behaviour and environmental effects.

## **8. Artistic Intent Statement**

Echo Garden brings participant movement to life within a digital meadow. Each visitor becomes a pollinator, providing visible and audible feedback on the direct impact of their actions. The exhibit focus on the importance of collaboration and attention in biodiversity, asking visitors to consider how their own behaviour affects ecosystems. At last, participants exits with a better understanding of their role in sustaining life on land.

## **9. Reflection and Learning Outcomes**

Working on it inspired our team to think in new ways about how to plan interactive experiences. We had learned how a digital ecosystem works and were beginning to appreciate the impact of what visitors do when they click around. We were working with people who could help us understand that our choices would matter. As we drew the storyboard and mapped the user flow, we could more easily translate our concepts into pictures. Creating the storyboard helped us understand how interactive design, storytelling and sustainability could enter the equation in a real-world project.